Research Statement of Zihan Yang

Introduction. Motivated by the limitations of how current robots operate, I aim to develop generalizable solutions that can adapt to various tasks and environments. I believe that endowing robots with generalizable decision-making capabilities is key to unlocking their transformative impact on society.

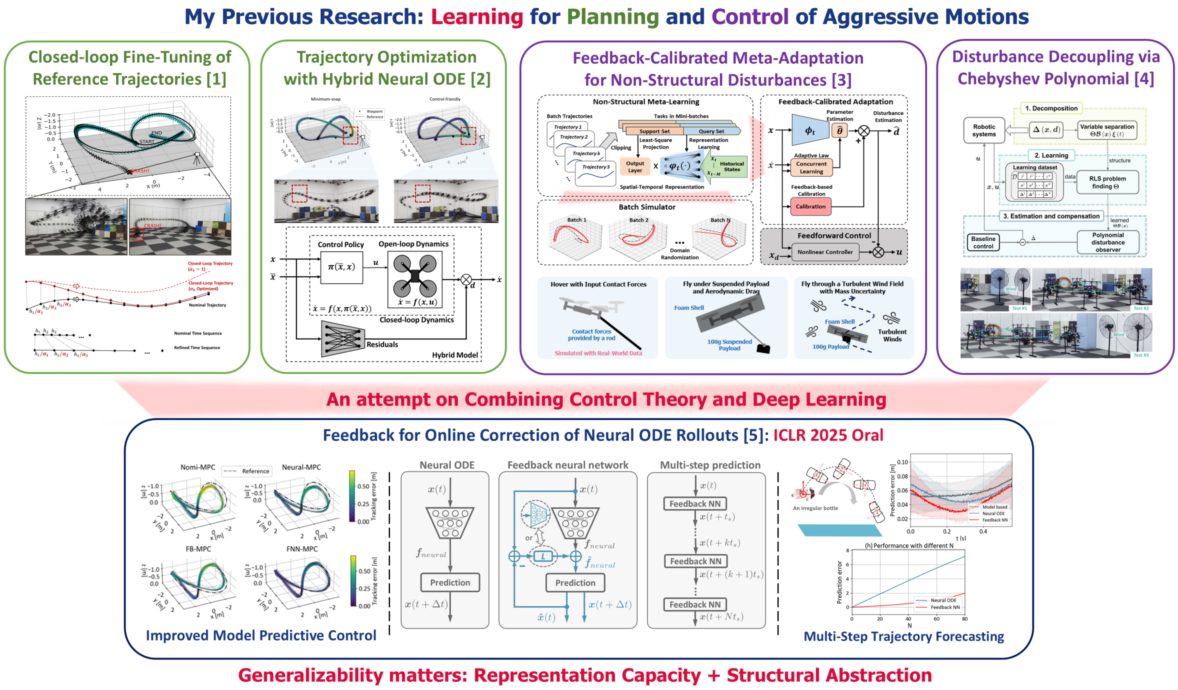

Figure 1: Overview of my previous research on learning-based planning and control, with extensions toward applying control theory to the design of deep learning models.

Previous Research. My research has focused on learning-based methods to compensate for residual physics and disturbances in robotic dynamics, particularly under aggressive motions. I have developed several frameworks from both planning and control perspectives. On the planning side, my work involves fine-tuning reference trajectories [1] to minimize control errors and leveraging neural ordinary differential equations (neural ODEs) to model and mitigate residual effects along optimized trajectories [2]. For control, I have explored online learning of disturbances using meta-learned representations that enable real-time adaptation to novel disturbances [3]. Additionally, I have investigated the decoupling of disturbances into state-dependent and time-varying components using Chebyshev polynomials [4]. Through these efforts, I observed that model generalizability stems from both **representation capacity** and **structured abstraction**, with the latter benefiting significantly from principles in control theory. Building on this insight, I proposed a feedback-augmented neural ODE framework that learns latent dynamics while incorporating online state feedback to calibrate model rollouts, resulting in more generalizable online predictions [5]. Despite promising results, these models are not yet directly applicable to more complex tasks, such as locomotion of articulated robots, long-horizon planning, or decision-making in manipulation. This motivates my ongoing efforts further to enhance the generalizability and scalability of learning-based robotic systems.

Current Interests. Generative models, such as diffusion-based [6], [7] and flow-based architectures [8], have demonstrated remarkable compositional generalization. When conditioned on specific tasks or environments, these models exhibit strong adaptability in robotic applications [9], [10], [11]. However, integrating generative models into low-level control frameworks remains challenging due to high inference latency and the need to enforce soft dynamic feasibility constraints. To address these challenges, I am working on approaches to embed generative models within model predictive control (MPC) frameworks, while ensuring that the overall computational complexity remains compatible with real-time MPC requirements.

Future Research. My future research will focus on learning generalizable solutions for robotic applications. Optimal control (OC) and reinforcement learning (RL) are two promising approaches to solve control, planning, and higher-level decision-making problems in robotics[12]. Enhancing model generalization is crucial across both OC and RL frameworks, while policy generalization is uniquely critical in RL settings. To address these challenges, I plan to pursue two interrelated directions: (i) designing structural abstractions that enable adaptable representations across diverse environments, tasks, and robot morphologies; and (ii) leveraging capable and expressive models, such as generative world models, to capture rich task and environment variations and support scalable decision-making. Orthogonal to the above two directions, I am also interested in fine-

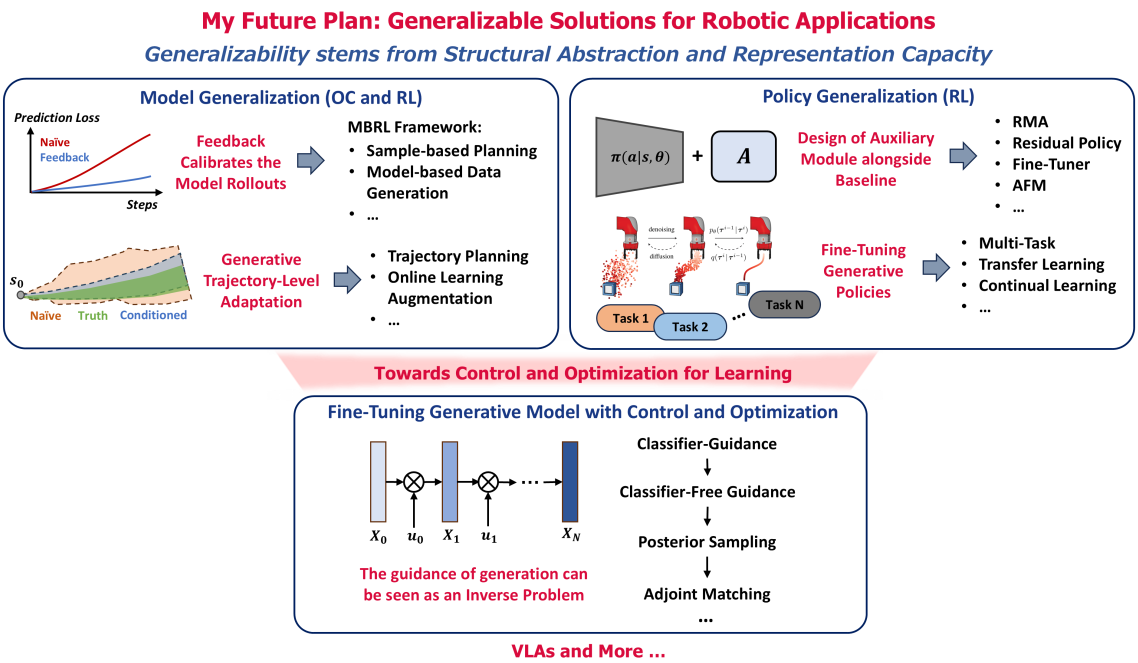

Figure 2: Research roadmap illustrating my future work on generalizable modeling in robotics, with extensions toward control-guided generative models and advanced embodied intelligence through vision-language-action frameworks.

tuning generative models using control and optimization. The guidance of diffusion-like generative models can be seen as an inverse problem [13], [14], where a goal condition or posterior distribution is desired to be achieved. This opens up the new possibility of integrating generative models with control and optimization. In addition, exploring the integration of more advanced embodied foundation models, such as Vision-Language-Action Models [15], represents a promising direction. These models combine large-scale perception, reasoning, and decision-making capabilities, potentially enabling robots to perform a broader range of tasks with greater autonomy.

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